Customer and Product Profitability Analysis

A deep dive into profit concentration, customer value, and product-level losses

795 Customers	**3,788** Products	**427** Customers for 80% profit	**67** Loss-making customers	**3** Segments

Executive Summary

This analysis extends the SuperStore sales performance work by focusing on customer and product profitability. The key finding is that profit is meaningful across a broad customer base rather than being dominated by only a few accounts. At the same time, losses are clearly concentrated in a smaller group of customers and specific products, especially 3D printers, machines, and tables.

Business question	Which customers and products create the most profit, which ones destroy margin, and how concentrated is profit across the customer base?
Scope	795 customers, 3,788 products, 2011–2014
Main customer finding	427 of 728 profitable customers, about 59%, generate 80% of positive customer profit
Main product finding	Losses are concentrated in selected Technology Machines and Furniture Tables products
Main recommendation	Manage profitability at customer and product level, not by sales volume alone

Prepared as a data analytics portfolio case study.

1. Customer Profit Concentration

The customer base contains 795 unique customers. Of these, 728 customers are profitable and 67 customers are loss-making. The top 427 profitable customers generate 80% of positive customer profit. This means that about 59% of profitable customers, roughly 60%, are needed to reach 80% of positive profit. This is a relatively broad profit base, which means the business is not dependent on only a handful of accounts.

ABC classes are based on cumulative positive customer profit. Class A customers generate the first 80% of positive profit, Class B customers bring cumulative profit from 80% to 95%, and Class C customers complete the remaining profitable customer base. Loss customers have negative total profit and are separated from the profitable ABC groups.

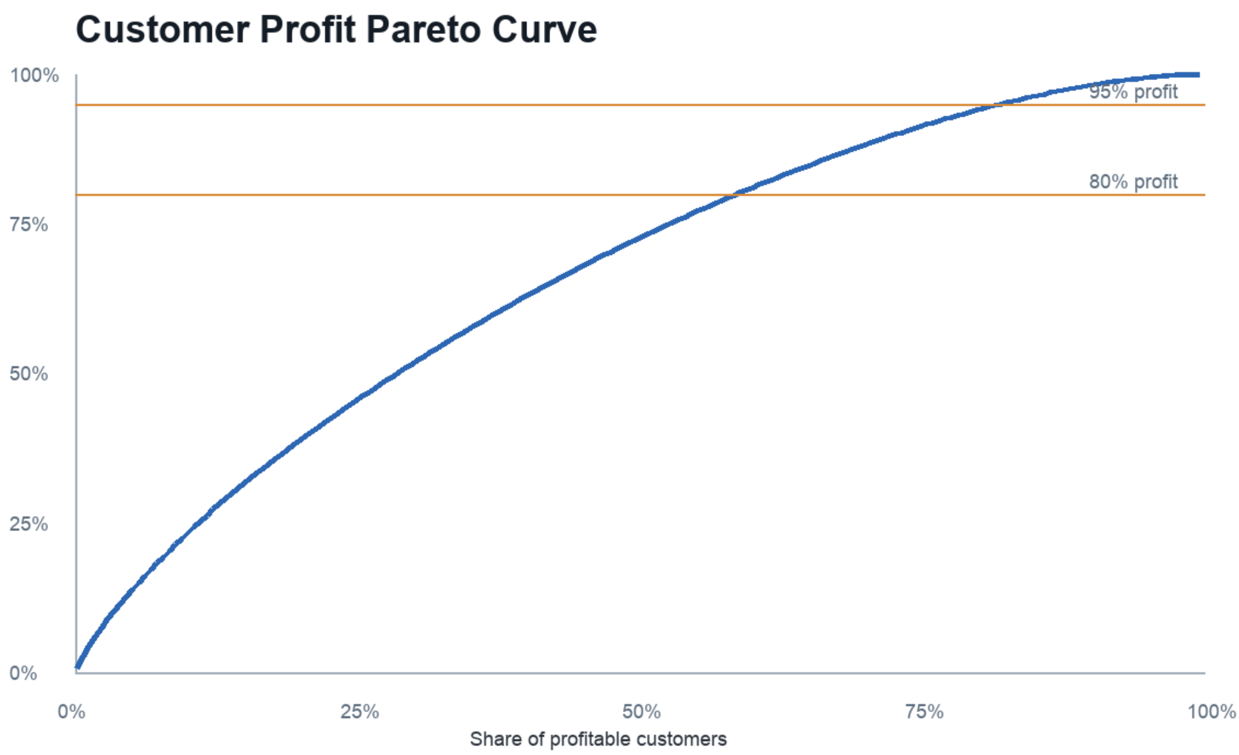

Figure 1: Customer profit Pareto curve

Class	Customers	Sales	Profit
A	427	$7,589,112	$1,227,368
B	169	$2,343,587	$229,796
C	132	$1,761,355	$77,384
Loss	67	$948,851	-$65,513

2. Most Profitable Customers

The most profitable customers combine meaningful sales volume with strong margin performance. They are not simply the largest buyers; they also generate attractive profit per dollar of sales.

Figure 2: Top customers by profit

Customer	Profit	Sales	Profit margin
Tamara Chand	$8,673	$37,453	23.2%
Raymond Buch	$8,453	$29,601	28.6%
Sanjit Chand	$8,205	$26,524	30.9%
Hunter Lopez	$7,817	$30,246	25.8%
Bill Eplett	$7,410	$28,480	26.0%

3. Loss-Making Customers

There are 67 customers with negative total profit. These customers are not automatically bad accounts, but they should be reviewed before additional discounts or promotions are expanded to them. The review should focus on discounting, product mix, shipping economics, and whether losses are isolated or recurring.

Customer	Profit	Sales	Profit margin
Cindy Stewart	-$6,152	$13,432	-45.8%
Luke Foster	-$3,644	$13,422	-27.2%
Grant Thornton	-$3,578	$20,226	-17.7%
Candace McMahon	-$2,799	$14,798	-18.9%
Skye Norling	-$2,638	$14,225	-18.5%

4. Product Profitability

The strongest products are mostly Technology items, especially copiers, phones, and routers. These products combine scale with strong profit margins.

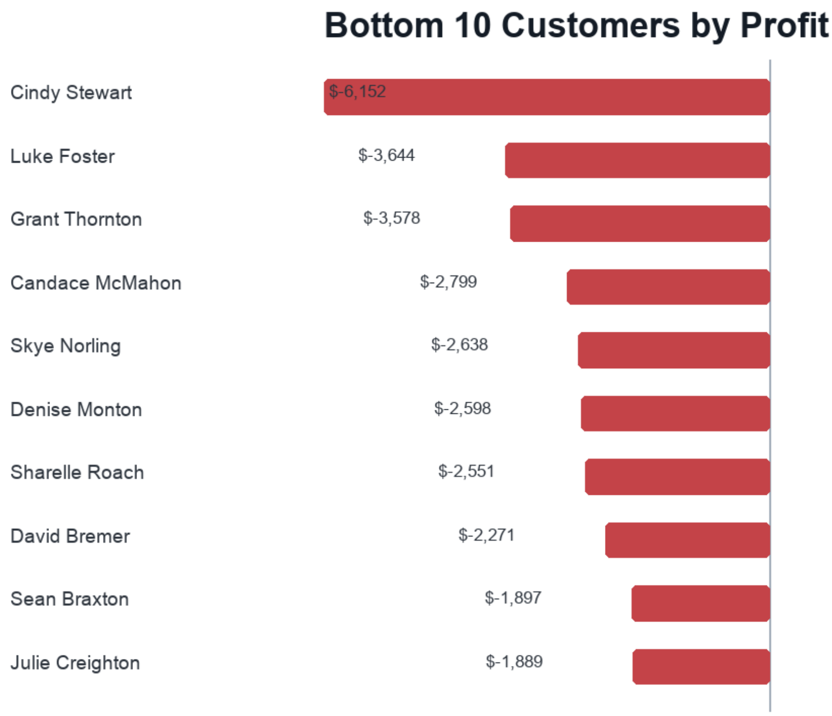

Figure 3: Bottom customers by profit

Product	Profit	Sales	Profit margin
Canon imageCLASS 2200 Advanced Copier	$25,200	$61,600	40.9%
Cisco Smart Phone, Full Size	$17,239	$76,441	22.6%
Motorola Smart Phone, Full Size	$17,027	$73,159	23.3%
Hoover Stove, Red	$11,808	$31,664	37.3%
Sauder Classic Bookcase, Traditional	$10,672	$39,110	27.3%

5. Product Losses

The largest product losses are concentrated in selected Technology Machines and Furniture Tables products. This aligns with the earlier category-level finding that Tables reduce Furniture profitability. High average discounts are a common pattern among the weakest products.

Product	Profit	Sales	Profit margin
Cubify CubeX 3D Printer Double Head Print	-$8,880	$11,100	-80.0%
Lexmark MX611dhe Monochrome Laser Printer	-$4,590	$16,830	-27.3%
Motorola Smart Phone, Cordless	-$4,447	$38,935	-11.4%
Cubify CubeX 3D Printer Triple Head Print	-$3,840	$8,000	-48.0%
Bevis Round Table, Adjustable Height	-$3,650	$5,655	-64.5%

6. Segment View

Customer segment is less important than product mix and discount behavior. Consumer, Corporate, and Home Office all show similar profit margins around 11.5–12.0%.

Figure 4: Top products by profit

Segment	Sales	Profit	Profit margin
Consumer	$6,508,141	$749,240	11.5%
Corporate	$3,824,808	$442,786	11.6%
Home Office	$2,309,956	$277,009	12.0%

Recommendations

- Protect high-profit customers with targeted retention and service quality.
- Review the 67 loss-making customers before expanding discounts or promotions to them.
- Investigate high-loss products, especially 3D printers, machines, and tables.
- Use customer profitability, not sales alone, as the primary account-prioritization metric.
- Keep segment-level strategy broad, but manage discounting and product mix at customer/product level.

Appendix: Reproducibility

The analysis can be reproduced with:

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python scripts/customer_product_analysis.py
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Figure 5: Bottom products by profit